

Entropy-Guided Dataset Reduction for Machine-Learned Interatomic Potentials

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Abstract

Machine-learned interatomic potentials (MLIPs) deliver near-quantum-mechanical accuracy at a fraction of the computational cost. Their performance depends on the quality and composition of the training data: the training set must cover the range of atomic environments the model will encounter, yet unnecessary redundancy increases training cost without improving accuracy and can inadvertently introduce training biases. Dataset pruning (i.e., selecting a compact, representative subset from a larger collection) has emerged as a practical strategy for building high-quality training sets. Recently, methods based on informationtheoretic diversity measures such as QUESTS (Quick Uncertainty and Entropy from STructural Similarity) have shown significant promise, however, QUESTS requires specification of a lengthscale parameter (kernel bandwidth) that controls how similar two atomic environments must be before one is considered redundant. We show that operationally useful values of this parameter vary across structurally distinct phases, making naive automated pruning of heterogeneous datasets challenging. We present a pruning framework that addresses this problem by adapting the bandwidth independently for distinct thermodynamic subsets of the training dataset using an automatable elbow criterion applied to QUESTS informationentropy metrics. Farthest-point sampling (FPS) is then applied within each group to select maximally diverse atomic environments. Using the Small Carbon ChIMES 2.0 dataset comprising diamond, graphite, and liquid carbon spanning a broad range of thermodynamic conditions, we show that use of a single global bandwidth yields pruned datasets that can be strongly dominated by a single phase, particularly when the distribution of environments varies significantly between phases (e.g., liquid vs crystalline).^{1,2} Relative to global pruning at the same retained fraction, group-aware adaptive pruning reduces test force-MAE by up to 37% for diamond and 26% for graphite for models trained on only 5% of the original training data, with corresponding improvements in radial distribution function (RDF) fidelity. Benefits are smaller in structurally

heterogeneous conditions where a single thermodynamic label encompasses multiple distinct local bonding environments with different characteristic length scales, indicating that finer grouping is the natural remedy in such regimes. Across crystalline conditions, the elbow-selected bandwidth aligns with the onset of rapid error growth as the dataset is pruned, providing a practical, model-agnostic guide for pruning decisions that requires no model training.

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broad range of thermodynamic conditions, we show that use of a single global bandwidth yields pruned datasets that can be strongly dominated by a single phase, particularly when the distribution of environments varies significantly between phases (e.g., liquid vs crystalline).^{1,2} Relative to global pruning at the same retained fraction, group-aware adaptive pruning reduces test force-MAE by up to 37% for diamond and 26% for graphite for models trained on only 5% of the original training data, with corresponding improvements in radial distribution function (RDF) fidelity. Benefits are smaller in structurally heterogeneous conditions where a single thermodynamic label encompasses multiple distinct local bonding environments with different characteristic length scales, indicating that finer grouping is the natural remedy in such regimes. Across crystalline conditions, the elbow-selected bandwidth aligns with the onset of rapid error growth as the dataset is pruned, providing a practical, model-agnostic guide for pruning decisions that requires no model training.

1 Introduction

The development of machine-learned interatomic potentials (MLIPs) has significantly expanded the scope of problems accessible to computational materials science, enabling simulations orders of magnitude faster than density functional theory (DFT) while retaining near-quantum mechanical accuracy.^{3–8} By learning to approximate the potential energy surface (PES) obtained from first-principles calculations, MLIPs enable molecular

dynamics (MD) simulations over spatial and temporal regimes that remain computationally prohibitive for *ab initio* methods.^{9,10} This capability has driven adoption across a range of applications, from catalysis and energy storage to prediction of phase transitions and mechanical properties.^{11–14}

Most MLIPs rest on the assumption that the total potential energy of a system decomposes into contributions associated with individual atoms or clusters of atoms, each of which depends only on the arrangement of neighbors within a finite cutoff radius. The local environment of a given atom is encoded through a descriptor^{15,16} that is ideally invariant to translation, rotation, and permutation of like atoms; common choices include atom-centered symmetry functions (ACSFs),⁴ smooth overlap of atomic positions (SOAP),¹⁵ and permutation-invariant polynomial expansions.^{8,17} A regression framework then maps descriptors to atomic energies, with forces and stresses obtained analytically as derivatives of the total energy with respect to atomic positions. The regressor may be linear in a set of basis functions, as in ChIMES (Chebyshev Interaction Model for Efficient Simulation)¹⁷ and ACE (atomic cluster expansion)⁸ interatomic potentials, or nonlinear, as in neural-network and message-passing architectures,^{4,9,10,14} where the representation of the local environment is learned rather than fixed in advance. Parameters are determined by minimizing a loss function that penalizes deviation of predicted energies, forces, and stresses from reference values computed for a set of atomic configurations, i.e., the training set. The pruning framework developed here operates on atomic-environment descriptors and is therefore independent of the architecture ultimately trained; we prune using QUESTS descriptors and fit graph neural network models.

Although many factors influence MLIP accuracy, including model capacity and the optimization protocol, three are particularly consequential: the expressiveness of the descriptor and regressor, the fidelity of the reference method used to label configurations, and the composition of the training set. The first two are design decisions made once and are well characterized in the literature.^{6,13} The third, by contrast, evolves throughout dataset construction, and quantifying which environments are information-rich remains difficult. Because an MLIP is an interpolative model, its predictions are most reliable for atomic environments well represented in the data on which it was trained; uneven sampling and accumulated redundancy therefore

inflate labeling and training costs while weighting the fit toward over-represented motifs.^{18,19} Training set construction is therefore a practical bottleneck in MLIP development, and it is the focus of this work.

Training data are typically generated by computing energies and forces with methods like DFT for atomic configurations obtained from a variety of sources, including DFT-MD or DFT-Monte Carlo (MC) simulations,^{20,21} existing crystal structure databases,²² random structure searches,²³ and high-throughput discovery pipelines.^{24,25} Each source imposes its own sampling characteristics; configurations drawn directly from crystal structure databases, for instance, can introduce significant bias into the training set, as equilibrium structures may be overrepresented relative to transition states and defect environments that control kinetic processes. To address this limitation, active learning strategies²⁶ have gained prominence as a systematic approach to training set construction.^{27–29} In active learning frameworks, an initial model is trained on a seed dataset, then used to drive MD or MC simulations that generate new candidate configurations. Each candidate is evaluated for informativeness using an uncertainty metric (e.g., ensemble disagreement,^{27,30} potential information gain,²⁸ or Bayesian uncertainty estimates^{29,31}); the most informative candidates are selected for DFT relabeling and added to the training set. This iterative process continues until convergence criteria are met, typically based on model performance on held-out validation data or uncertainty scores falling below specified thresholds.

However, any large dataset, whether generated by active learning or direct simulation sampling, can accumulate redundancy by way of configurations that are structurally similar to others already in the set and therefore contribute little new information.^{18,32} In active learning, this can occur when uncertainty estimates are noisy and select marginally informative configurations,^{33,34} or when the iterative protocol over-samples difficult-to-predict regions that are rare under typical simulation conditions.^{19,35,36} Regardless of origin, redundant configurations increase training cost without improving model performance and can bias the model, e.g., toward accurate description of over-represented configurations at the cost of those less well-represented in the training set. Identifying and removing redundancy serves two complementary purposes: pruning a large dataset prior to training reduces computational overhead and cura-

tion effort, while integrating redundancy detection into automated workflows such as active learning minimizes cumulative labeling costs and accelerates convergence toward a compact, high-quality training set.^{37,38} Pruning removes near-duplicates while preserving coverage, which for a fixed labeling budget moves the training set toward broad, approximately uniform sampling of the relevant descriptor space.

1.1 Core Datasets for Machine Learning Interatomic Potentials

Given a large collection of configurations (whether generated by active learning, direct MD sampling, or database extraction), a natural question is whether the full set is necessary for accurate modeling or whether a compact subset can match its predictive performance. Pruning to such a subset reduces the cost of model training and retraining, accelerates hyperparameter search, and simplifies dataset maintenance, while the original large set remains available if needed. Such pruning is especially valuable when a dataset is to be relabeled with a more expensive quantum-mechanical (QM) method (e.g., a higher-rung functional) whose cost is prohibitive over the full training set; a compact subset that preserves predictive performance can be relabeled at the higher level of theory and used to train the ML model, recovering the accuracy of the expensive method at a fraction of the labeling cost. Selection here operates on atomic environments, not whole structures. Deciding which complete configurations to recompute at the higher level of theory is then a separate, non-trivial step from the pruning addressed here.

Core dataset construction addresses the underlying question: given a large collection of atomic environments with known energies and forces, how can we identify a compact subset D_{core} of the full dataset D_{full} that yields models at least as accurate as a model trained on D_{full} ? We frame the target as “at least as accurate” rather than “equally accurate” because D_{full} may itself contain redundancy and uneven sampling that bias the fit, so it is not guaranteed to be the optimal training set.

Mathematically, this is an optimization problem:

$$\min |D_{\text{core}}| \text{ s.t. } f(M(D_{\text{core}})) \geq f(M(D_{\text{full}})) - \epsilon, \quad (1)$$

where $D_{\text{core}} \subseteq D_{\text{full}}$, $M(\cdot)$ denotes the fitted model, $f(\cdot)$ measures model quality, and $\epsilon \geq 0$ is a user-

specified tolerance on the performance loss deemed acceptable relative to the full-data reference.

The theoretical foundation for training set pruning lies in the observation that the information content of atomic configuration datasets is not uniformly distributed. Some local atomic environments provide insight into regions of the potential energy surface that are not yet well-represented, while others closely resemble environments already in the set and contribute minimal new information. This concept is formalized through the notion of dataset efficiency, defined as the ratio of the dataset’s Shannon (information) entropy to its theoretical maximum for a dataset of that size, i.e., $\eta = \mathcal{H}(h)/\mathcal{H}_{\text{max}}$, where $\mathcal{H}_{\text{max}} = \log(n)$ and n is the dataset size.³⁹ Intuitively, a dataset with high efficiency covers descriptor space broadly and uniformly relative to its size: each environment occupies a distinct region and the actual diversity approaches the theoretical maximum. Conversely, a dataset with low efficiency contains significant redundancy, with many environments clustering tightly in descriptor space such that adding more provides diminishing informational returns. While this metric motivates pruning to eliminate redundancy within a dataset, optimizing efficiency alone introduces an edge case: in the limit of a single retained configuration, efficiency is trivially maximal, yet the set trains nothing. To avoid this artifact, efficiency computations should be subject to a minimum dataset size constraint.

Several approaches to core dataset construction have been explored in the MLIP literature. For example, distance-based clustering methods group environments using structural similarity metrics, then select representative environments from each cluster,^{35,40,41} though these approaches are sensitive to the choice of distance metric and to the length scale at which similarity is assessed. Cersonsky et al.⁴² developed principled subselection using eigenvector-based methods that account for descriptor covariance structure. Complementary work has used unsupervised clustering to learn training-set weights within discovered groups, rather than discarding configurations, while jointly optimizing model hyperparameters for property-targeted “bespoke” potentials.⁴³ These methods provide theoretical guarantees about coverage but still require a choice of metric and associated hyperparameters in descriptor spaces that typically span tens to hundreds of dimensions for modern MLIPs. Energy-based selection methods prioritize configurations with high potential energies or un-

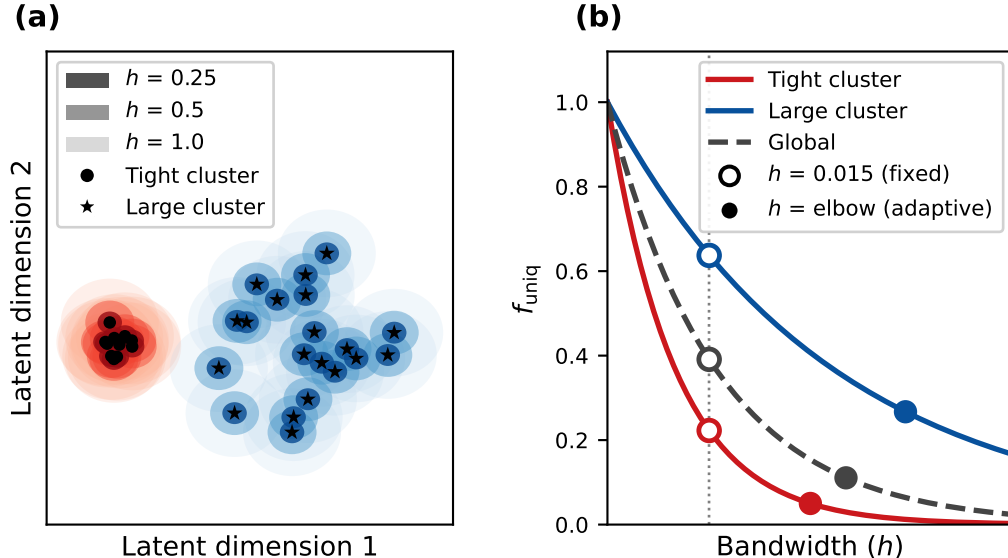


Figure 1: Conceptual illustration of bandwidth sensitivity and the motivation for adaptive, per-group bandwidth selection. (a) A schematic two-dimensional descriptor (latent) space with a tight cluster (red dots) and a large cluster (blue stars). Gaussian kernel contours at three bandwidths ($h = 0.25, 0.5, 1.0$) illustrate how a single global bandwidth simultaneously over-broadens the tight cluster (merging distinct environments) and under-broadens the large cluster (treating similar environments as unique). (b) The unique fraction $f_{\text{uniq}}(h) = \mathcal{H}(h)/\mathcal{H}_{\text{max}}$ (formally defined in Sec. 2.1) as a function of bandwidth h for the tight cluster (red), the large cluster (blue), and the pooled dataset without stratification (gray dashed, “Global”). Open circles mark the fixed bandwidth ($h = 0.015$); filled circles mark the elbow-selected (adaptive) bandwidth for each curve. A single fixed bandwidth is either too narrow to resolve redundancy in the large cluster or too broad to preserve structure in the tight cluster, so the Global curve at $h = 0.015$ reports a substantially higher f_{uniq} for the tight cluster than its own elbow-selected value. Assigning each group its own bandwidth, calibrated to its characteristic length scale, removes that mismatch; the elbow is an operational choice rather than a provably optimal one.

usual bonding patterns,^{27,29,44} operating under the assumption that rare events provide more information than common structures. However, this approach can bias datasets toward high-energy, potentially unphysical configurations that may not improve model performance for typical simulation conditions. All distance-based selection strategies, including the farthest point sampling used here, depend on how distance is defined in descriptor space and on the length-scale hyperparameters that determine that metric. Calibrating these length scales becomes increasingly difficult in high dimensions. The procedure below addresses this calibration for entropy-based scoring and could, in principle, be combined with other distance-based selectors.

Recent work has introduced quantitative measures of dataset quality through information-entropy-based calculations.^{38,39} This approach provides a framework for comparing pruning strategies and optimizing dataset composition. In QUESTS specifically, structural diversity is quantified through the Shannon entropy of a kernel

density estimate over atomic-environment descriptors, and the bandwidth h of that kernel sets the length scale at which two environments are treated as redundant (see Fig. 1). Because different material phases exhibit different structures and characteristic fluctuations about those structures, a single global h is unlikely to be appropriate across heterogeneous datasets and may limit the effectiveness and transferability of entropy-guided pruning to new materials systems. This difficulty in choosing h therefore convolutes both the interpretation of efficiency as a quality measure and the goal of fully automating dataset optimization. Efficiency is meant to reward diversity, with high values signaling broad, uniform coverage of descriptor space. When the bandwidth is set too small, however, every environment appears distinct from its neighbors, and even a highly redundant dataset registers as near-maximally efficient. Efficiency then no longer means what it was intended to. In this work, we address these challenges through a pruning framework that adapts to structural differences between groups of atomic environments

defined by thermodynamic state point.

The remainder of this paper is organized as follows. Section 2.1 defines the QUESTS entropy and efficiency metrics and the role of the bandwidth parameter. Section 2.2 describes the state-point grouping, the elbow-based bandwidth selection criterion, and the farthest-point sampling protocol applied within each group. Section 3 evaluates global and group-aware pruning on the Small Carbon ChIMES 2.0 dataset through force errors and radial distribution functions across diamond, graphite, and liquid conditions. Section 4 summarizes the findings and their implications for automated training set curation.

2 Methods and Computational Details

2.1 QUESTS

Our framework employs the QUESTS methodology^{38,39,45} as the foundation for quantifying dataset information content and guiding core dataset construction. QUESTS provides a model-agnostic approach to uncertainty quantification and dataset analysis specifically designed for atomistic simulation data. Unlike traditional machine learning metrics that require pretrained models, QUESTS leverages information-theoretic principles to assess dataset quality, structural diversity, and learning convergence directly from atomic environments.

In the QUESTS framework, the information content of a group of atomic environments is quantified using the Shannon entropy, providing a direct measure of structural diversity within the dataset. The entropy formulation builds upon the fundamental information-theoretic relationship:⁴⁶

$$\mathcal{H} = - \int p(x) \log p(x) dx, \quad (2)$$

where $\mathcal{H}$ is the (differential) Shannon entropy, $p(x)$ is the probability density function over the descriptor space, and x denotes a point in that space, i.e., a realization of the atomic-environment descriptor.

For practical implementation, kernel density estimation (KDE) approximates the underlying probability distribution $p(x)$, enabling entropy calculations.⁴⁷

$$\mathcal{H}(\{\mathbf{X}\}) = -\frac{1}{n} \sum_{i=1}^n \log \left[\frac{1}{n} \sum_{j=1}^n K_h(\mathbf{X}_i, \mathbf{X}_j) \right] \quad (3)$$

where K_h is the Gaussian kernel, described in greater detail below. n is the number of atomic environments in the set $\{\mathbf{X}\}$ being evaluated (e.g., a single state-point group or the full dataset), and $\mathbf{X}_i, \mathbf{X}_j$ are the QUESTS descriptor vectors representing atomic environments i and j . The neighbor distances, and therefore the computed entropy, depend on the user’s choice of $\mathbf{X}$. Here, we use the default QUESTS descriptor, in which each atomic environment is represented by a sorted vector of inter-atomic distances within the neighborhood of the 32 nearest neighbors of a central atom. This vector includes distances between the neighbors themselves in addition to distances from the central atom, so that three-body information is incorporated; further detail on its construction is given in Ref. 39. Other representations may be substituted within the same workflow if preferred.

The Gaussian kernel employed by the KDE is given below:

$$K_h(\mathbf{X}_i, \mathbf{X}_j) = \exp \left(\frac{-\|\mathbf{X}_i - \mathbf{X}_j\|^2}{2h^2} \right), \quad (4)$$

where h is the bandwidth parameter controlling resolution of structural similarity. The original QUESTS framework established $h = 0.015$ as a default, universal fixed bandwidth, which we adopt as the reference value for fixed-bandwidth calculations in this work. The bandwidth parameter influences entropy calculations by controlling the broadness of the density estimate. While this parameter dependence limits the absolute interpretation of entropy measures, consistent bandwidth selection enables meaningful relative comparisons between datasets and structural motifs.

The choice of bandwidth h will significantly impact results. A fixed dataset can be driven to almost any efficiency value by modulating h ; at the narrow (small h) extreme, the kernel cannot detect overlap between structurally similar environments. Hence, every environment appears unique regardless of actual redundancy, and efficiency approaches unity at any dataset size. At the broad (large h) extreme, distinct environments are considered redundant and efficiency approaches zero. To avoid this misleading interpretation, we henceforth rename efficiency to $f_{\text{uniq}}(h)$, the fraction of configurations that are classified as unique for a given bandwidth, h .

2.2 Entropy-Guided Pruning

The Small Carbon ChIMES 2.0 dataset is distributed with labels identifying the 14 (temperature, density, phase) state points from which its configurations originated, as described in the Supporting Information (Sec. S1, Table S1). We use these labels directly to partition atomic environments into groups. No clustering is performed on the descriptors themselves, so the grouping used throughout this work is a physically motivated prior supplied with the dataset rather than something inferred from the data. Within each group, we then compute localized entropy measures and apply a selection protocol that retains representative atomic environments while accommodating the structural heterogeneity characteristic of comprehensive MLIP training datasets.

Within each group, we compute the distribution of pairwise distances between all atomic environments in QUESTS descriptor space. The mean and standard deviation of this distribution provide an empirical estimate of the characteristic length scale over which environments in that group differ, and set the scale on which the bandwidth must operate if the kernel is to resolve real structural variation. Consistent with the coordination-shell argument of Sec. 2.1, liquid groups yield broader distance distributions than crystalline groups, so the required bandwidths can be anticipated in advance of any entropy calculation.

We compare three approaches to bandwidth selection, differing in the degree to which they adapt to group-level structure. All three approaches select environments by farthest point sampling (FPS), which sequentially retains the environment farthest in QUESTS descriptor space from all previously selected environments, starting from a random seed. This greedy procedure maximizes the minimum pairwise distance in QUESTS descriptor space across the retained set, and the approaches differ in the scope over which it is applied and in the bandwidth used to score the result.

1. **Global-Fixed:** fixed bandwidth, global sampling (default QUESTS strategy)
2. **Stratified-Fixed:** fixed bandwidth, group-stratified sampling
3. **Stratified-Adaptive:** adaptive bandwidth, group-stratified sampling

Global-Fixed applies a single, globally-defined bandwidth h_{global} to the entire dataset and performs FPS directly on the full configuration space

without regard to group membership, consistent with the default application of QUESTS described in Schwalbe-Koda et al.³⁹. We include this method as a reference point against which group-stratified approaches are compared. The global bandwidth follows the default value established in the original QUESTS framework, $h = 0.015$, rather than being tuned to the statistics of any particular dataset: Schwalbe-Koda et al.³⁹ selected this value because it corresponds approximately to the descriptor-space distance between two FCC environments ($k = 32$ nearest neighbors, $r_{\text{cut}} = 5 \text{ \AA}$) at the equilibrium lattice parameter ($3.58 \text{ \AA}$) and the same environment with unit-cell parameters rescaled by 1%. Because selection is driven by distances in the pooled descriptor space, this strategy retains proportionally more environments from diffuse regions of that space (e.g., disordered liquid phases) than from tightly distributed regions (e.g., crystalline phases). Whether that allocation is desirable is not obvious *a priori*. Diffuse regions do carry more information per configuration at a given bandwidth, and concentrating the retained budget there may be the correct choice for some applications; for others it may under-represent crystalline motifs that matter for the target properties. We therefore treat the allocation produced by global FPS and the balanced allocation produced by stratified sampling as competing hypotheses, and compare them empirically rather than presuming either to be unbiased.

Stratified-Fixed employs the same fixed bandwidth h_{global} but performs FPS independently within each group, thereby enforcing equal fractional retention from every state point instead of a single global ranking, while maintaining computational simplicity.

We include Stratified-Fixed to disentangle the contributions of group-stratified sampling and adaptive bandwidth selection: by holding bandwidth fixed while adopting stratified sampling, we isolate the effect of sampling strategy alone, independent of bandwidth adaptation. Atomic environments are sampled proportionally (or according to a specified allocation) from each group, so that every labeled structural motif is represented in proportion to its original size regardless of its relative entropy score. There is no *a priori* reason that proportional representation must be preferable to global ranking, and we pose it here as a hypothesis to be tested rather than as a correction. The fixed bandwidth, meanwhile, is expected to remain poorly matched to groups whose characteristic length scales differ substantially from

the global value. Entropy estimates within each group may then be systematically skewed: overestimated for tight groups (kernel too broad relative to the group’s spread) or underestimated for diffuse groups (kernel too narrow), reducing the effectiveness of entropy as a selection criterion.

Stratified-Adaptive, our proposed approach, addresses the limitations of fixed-bandwidth approaches by selecting the bandwidth independently for each group. For each group c , we define an empirical bandwidth h_c^* as the elbow of the QUESTS $f_{\text{uniq}}(h)$ curve (Fig. 1b), the point of maximum curvature where f_{uniq} transitions from rapid decline to a shallow, slowly varying tail. At this scale the kernel resolves structural similarity and exposes redundancy without aggressively merging structurally distinct environments. The curve is computed on the full, unpruned set of environments in the group by sweeping h , without FPS. Because crystalline phases have sharp, discrete coordination shells while disordered liquids have broad, continuous distributions, this bandwidth differs across phases, so a single global value is a poor match for heterogeneous datasets. Assigning each group its own h_c^* lets the entropy scores reflect local information content at a scale appropriate to each structural regime. FPS is then applied at the fixed value h_c^* to select subsets.

Stratified-Fixed uses stratified sampling with a fixed bandwidth, so it differs from Global-Fixed only in the sampling and from Stratified-Adaptive only in the bandwidth. Comparing across the three therefore shows how much each factor matters on its own. We use it this way in the bandwidth analysis below. The accuracy and RDF comparisons that follow focus on Global-Fixed and Stratified-Adaptive, and we bring in Stratified-Fixed only where it helps explain a difference.

For each approach, we prune iteratively by FPS, applied within each group for Stratified-Fixed and Stratified-Adaptive and globally for Global-Fixed, and evaluate model performance as a function of retained dataset fraction. The pruning process proceeds in systematic 5% increments of the original group size, where group size is defined as the total number of atomic environments across all frames in that group (number of frames $\times$ number of atoms per frame). At each pruning level, we extract the selected atomic environments, reconstruct their corresponding simulation frames (configurations), and train independent NequIP models using identical hyperparameters and training protocols. The held-out test set for each state point con-

sists of 100 configurations randomly sampled from the same DFT-MD trajectories used to construct the training data, but excluded from training and pruning. Model performance is assessed through both training and held-out test-set force mean absolute error (MAE), with supplementary structural validation via radial distribution function (RDF) comparison. The optimal retention level cannot be known *a priori*, so we use this dense sweep as a validation experiment to test whether the elbow criterion identifies a suitable level on its own. In practice, one would apply the elbow criterion directly.

2.3 ChIMES Dataset

We evaluate our method on the Small Carbon ChIMES 2.0 dataset,^{1,2} which spans diamond, graphite, and liquid carbon state points covering densities from 0.50 to 3.68 g/cc and temperatures from 300 to 8000 K. Configurations were generated through active-learning-assisted ChIMES workflows and labeled with DFT forces and energies; dataset composition details are summarized in the Supporting Information (Table S1). This diversity makes the dataset a useful test for whether entropy-guided pruning remains reliable across highly varied structural regimes, and its combination of structurally coherent crystalline state points with internally heterogeneous liquids lets us probe where group-aware pruning succeeds and where it does not.

2.4 NequIP Model

We use NequIP,^{48–50} an E(3)-equivariant graph neural network, as a representative modern MLIP architecture. NequIP was selected because its equivariant design is well suited to carbon systems with multiple distinct bonding environments, and its open-source codebase⁵¹ and documentation⁵² facilitate reproducibility. The choice of NequIP also decouples the evaluation from the dataset’s provenance. The Small Carbon ChIMES 2.0 dataset was assembled through active learning driven by ChIMES models, so configurations were retained on the basis of ChIMES uncertainty. Refitting ChIMES to pruned subsets would therefore entangle the effect of the pruning criterion with the criterion that generated the candidate pool. Fitting an architecture with different descriptors and a different functional form removes that coupling and tests whether the retained environments

are informative for MLIP training generally rather than for ChIMES specifically. It also means the descriptors used for pruning differ from those used by the trained model, so any performance gains additionally indicate that the selection transfers across representations.

A consistent single-species (C) architecture and optimization protocol were used across all pruning conditions to isolate dataset-selection effects. We used a cutoff radius of 5.0 Å, three interaction layers, angular channels up to $l_{\max} = 1$ with parity enabled, 64 features per layer, eight Bessel radial basis functions, and a two-layer radial MLP of width 64. The model included a ZBL short-range pair potential and fixed per-type energy scale and shift parameters.

Optimization used the Adam optimizer⁵³ with an initial learning rate of 10^{-2} and a ReduceLROnPlateau learning-rate scheduler (decay factor 0.6, patience 5, minimum learning rate 10^{-6}), early stopping (patience 20), and exponential moving average weight averaging with decay 0.999. A force-only loss function was used (energy coefficient zero), consistent with our force-error-driven evaluation metrics. Additional architecture details are provided in the Supporting Information (Sec. S2).

3 Results and Discussion

In this section, we evaluate whether adaptive, group-stratified entropy-guided pruning improves model fidelity at fixed retained fraction. We first assess model accuracy as a function of dataset retention using both training and held-out test force MAE. We then examine fixed- and elbow-selected bandwidth behavior to interpret those accuracy trends and to establish how pruning thresholds vary across phases. Finally, we validate structural fidelity through radial distribution functions (RDFs) and analyze phase-dependent limits of the approach.

3.1 Aggregate and Per-Statepoint Model Accuracy

Fig. 2 shows training and test force MAE as a function of dataset retention for representative state points. The aggregate “Full” panel shows that group-stratified pruning yields lower RMSE than global pruning across the retention range for each phase, with the largest differences observed for crystalline state points. For example, at 5% re-

tention, group-stratified MAE for 3.68 g/cc, 300 K diamond and 2.56 g/cc, 1500 K graphite is approximately 37% and 26% lower than that of corresponding Global-Fixed models, respectively. These improvements are consistent with global FPS allocating a larger share of the retained budget to liquid environments, which occupy more diffuse regions of descriptor space at $h = 0.015$.

For low-density liquid carbon (0.5 g/cc, 1000 K), the group-stratified method remains favorable but with smaller margins (approximately 3% test MAE improvement for models trained on 10% of the data). For high-density liquid carbon (2.0 g/cc, 7000 K), performance is near-parity or slightly worse for group-stratified pruning at some retention levels. We attribute this to internal structural heterogeneity within high-density liquid groups: configurations assigned the same thermodynamic label span multiple distinct local bonding environments (sp , sp^2 , sp^3), such that within-group FPS cannot resolve these subpopulations and the premise that one label corresponds to one characteristic length scale no longer holds. Global FPS, which samples across the full dataset simultaneously, inadvertently captures this sub-structure through global distance maximization. Expanded per-statepoint trends as a function of retained atom count (Supplementary Figs. S3 and S5) and retained fraction (Supplementary Figs. S4 and S6) confirm this phase dependence across the full dataset.

3.2 Bandwidth Sensitivity and Elbow Selection

Fig. 3 compares the $f_{\text{uniq}}(h)$ of each group computed with the default QUESTS bandwidth³⁹ ($h = 0.015$) and with elbow-selected adaptive bandwidths; these values are tabulated in Supplementary Tab. 2. A dashed black horizontal line marks the Global-Fixed value, $f_{\text{uniq}} = 0.884$, obtained by applying $h = 0.015$ to the pooled dataset without stratification. Note that the default bandwidth is applied uniformly to all groups and to the whole dataset; it is not an elbow selection.

The default bandwidth yields near-unity f_{uniq} for almost every group, which would imply that nearly every atomic environment is unique and little can be pruned without information loss. This is unlikely to be the case. These configurations were selected by ChIMES-driven active learning, which chooses on the basis of whole-structures informativeness, whereas f_{uniq} is evaluated per envi-

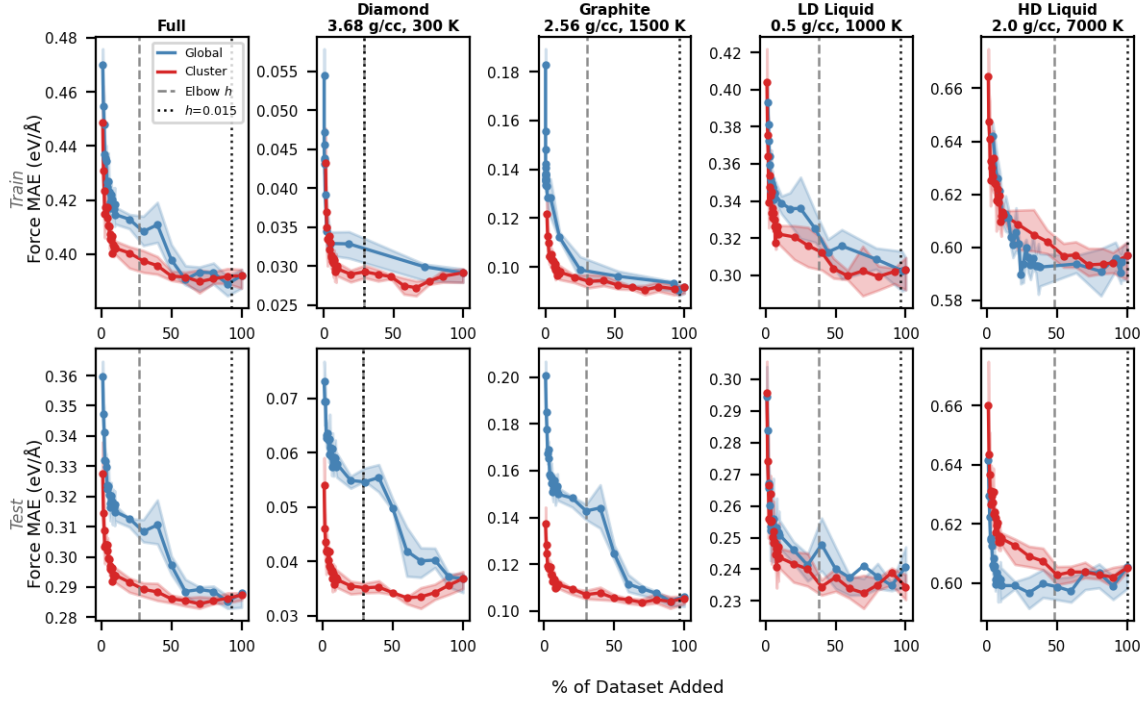


Figure 2: Training (top row) and held-out test-set (bottom row) force MAE versus amount of dataset used in training for representative state points. Blue denotes Global-Fixed (FPS applied globally, $h = 0.015$) and red denotes Stratified-Adaptive (FPS applied within each state-point group at its elbow-selected h_c^*); shaded regions indicate interquartile ranges (IQR) across independent runs. Dashed and dotted vertical reference lines indicate the retention fraction corresponding to the elbow-selected (adaptive) and $h = 0.015$ (fixed) bandwidths, respectively. In the Full aggregate panel, the elbow- h line corresponds to the elbow of the global (non-stratified) QUESTS $f_{\text{uniq}}(h)$ curve for the pooled dataset.

ronment. Since a single structure can contribute many near-equivalent local environments (e.g., in the case of a crystal), redundancy should be present at the environment level. At this bandwidth, however, the kernel is too narrow to register that redundancy and is therefore uninformative. Determining whether the environments are truly unique or merely unresolved requires an additional assessment, which is provided by the model comparisons in Sec. 3.1.

In contrast, elbow-selected bandwidths produce substantially lower f_{uniq} values across all groups, indicating significant within-group redundancy. Lower f_{uniq} values are consistent with the bandwidth resolving structural similarity between environments that the default narrow bandwidth cannot detect. No independent ground truth for environment similarity is available, so this interpretation rests on indirect evidence, i.e., subsets selected at the elbow bandwidth train better models at equal retention (Sec. 3.1).

Diamond (3.68 g/cc, 300 K) shows the closest agreement between the default and elbow-selected bandwidths ($h = 0.015$ and 0.017, respectively), re-

flecting small thermal structural fluctuations at low temperature that keep the group’s natural length scale close to the default. For these solid phases, differences in f_{uniq} between the two methods increase with temperature. Graphite shows this clearly: the f_{uniq} gap between methods is modest at 300 K (2.39 g/cc), widens at 1500 K (2.56 g/cc), and widens further at 3000 K (2.67 g/cc), corresponding to elbow-selected bandwidths of $h = 0.018$, 0.0347, and 0.0448, respectively. The trend follows from graphite’s weak interlayer (van der Waals) bonding, as interlayer sliding and stacking fluctuations broaden the descriptor-space spread well beyond what the fixed default bandwidth can resolve, so the default reports progressively higher f_{uniq} than the elbow-selected bandwidth, which adapts to track the growing spread. In contrast, the liquid state points show the divergence between default and elbow-selected bandwidths irrespective of temperature and density. The default bandwidth reports f_{uniq} values roughly 0.4 to 0.75 higher than the elbow-selected value across all seven liquid groups (e.g., 3.0 g/cc, 8000 K: $f_{\text{uniq}} = 0.95$ at $h = 0.015$ versus 0.21 at the elbow). This is

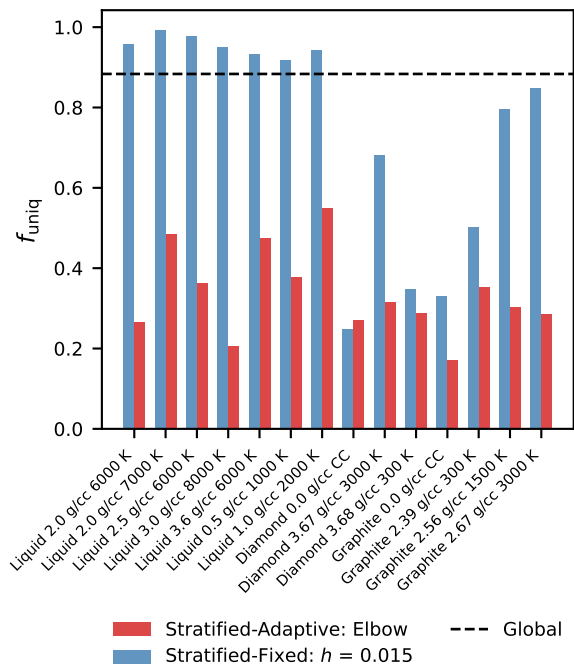


Figure 3: Comparison of QUESTS f_{uniq} computed with a fixed bandwidth ($h = 0.015$) versus an adaptive elbow-selected bandwidth for each thermodynamic state point in the Carbon ChIMES 2.0 dataset. The dashed black line (“Global”) marks the Global-Fixed reference ($f_{\text{uniq}} = 0.884$), obtained by applying $h = 0.015$ to the entire pooled dataset without stratification.

the diffuse-cluster limit illustrated in Fig. 1: liquids exhibit higher configurational entropy than solids, giving a large spread in atomic-descriptor space that the narrow default bandwidth cannot properly resolve. The full bandwidth- f_{uniq} trends underlying this comparison are shown in Supplementary Fig. S2 with the corresponding tabulated critical values in Table. S2.

3.3 Structural Validation via Radial Distribution Functions

Force errors quantify pointwise accuracy on the PES but do not establish that a pruned model reproduces the structure a simulation actually samples. For each pruned dataset, we trained a NequIP model and ran MD at every state point in the dataset, computing RDFs from the resulting trajectories using TRAVIS.^{54,55} Comparing each curve to that for the full-dataset model isolates if and how structural predictions change with training set composition and size. Each model was run as an NVT simulation in LAMMPS⁵⁶ using a 0.5 fs timestep for 10,000 steps at the correspond-

ing state-point temperature.

We find RDFs to be far less sensitive to dataset composition than force errors (Fig. 4). Where force MAE varies by tens of percent between pruning strategies at fixed retained fraction, RDF differences are small and, in several cases, within the trajectory-to-trajectory variation. The distinction follows from what each metric probes. An RDF is determined largely by the location of the PES minima and the curvature about them, so a model need only place equilibrium bond distances correctly and reproduce the local shape of the well to yield an accurate RDF. Force MAE is evaluated over the full test set, including configurations far from equilibrium that are visited rarely, if at all, in an equilibrium trajectory and therefore contribute negligibly to the RDF but are nevertheless important for accurate description of kinetic properties. A model can degrade substantially in its description of these environments while leaving the RDF nearly unchanged.

For diamond and graphite, group-pruned models reproduce peak positions and amplitudes somewhat better at low retained fractions and show lower RDF-RMSE across most of the pruning range. Phase-resolved results in the Supporting Information follow the same pattern, with the clearest separation for diamond and graphite (Supplementary Figs. S8 and S9), less for low-density liquid (Supplementary Fig. S10), and little to none for high-density liquid (Supplementary Fig. S11). The force-correlation analysis (Supplementary Fig. S7) likewise shows more favorable error-force trends for group-pruned models in the structurally coherent regimes.

The practical implication is that RDF agreement is a necessary but weak test of a pruned dataset. All pruning strategies considered here preserve equilibrium structure at retained fractions where force errors have already diverged appreciably, so structural validation alone would not distinguish between them. This is worth noting because RDFs are a common validation target for MLIPs; agreement there should not be taken as evidence that a training set has been pruned without consequence for the regions of the PES that matter for reactive or far-from-equilibrium simulation. It also indicates that a more discriminating observable would be a better evaluation target for pruning methodology, and identifying one suited to the intended application, for instance elastic constants, defect formation energies, or transport coefficients, is a worthwhile direction for future assessments of

this kind.

3.4 Limits of Group-Stratified Pruning

Recall that in this work, each group corresponds to a unique combination of phase, density, and temperature (e.g., diamond at 3.68 g/cc and 300 K is one group; liquid at 2.0 g/cc and 7000 K is another). The high-density liquid (i.e., the one regime where stratified pruning does not outperform global) suggests that group definition also affects selection efficacy. In this regime, a single thermodynamic label can encompass multiple local bonding motifs ($sp/sp^2/sp^3$), so within-group FPS spreads its selections across internally distinct subpopulations without resolving each one well. This internal heterogeneity is visible in the UMAP⁵⁷ projection of descriptor-space distributions (Supplementary Fig. S1), where high-density liquid groups show internal sub-clustering that thermodynamic labels do not separate. Since these subpopulations are far apart in descriptor space, global FPS selects from each of them without needing the group labels and is thereby able to match the stratified pruning result.

Several limitations of relying on predefined labels follow from this: (1) while labels are available and largely meaningful for this dataset, that is not generally the case; (2) per-label stratification does not address redundancy shared across state points; (3) the grouping is supplied rather than discovered, so the approach as implemented is not fully automatic. Generating groups via unsupervised clustering provides a label-free strategy for overcoming these limitations worth exploring in future work.

Across crystalline state points, elbow-derived retention markers in Fig. 2 and Fig. 4 align reasonably with the onset of rapid error growth, indicating that the elbow criterion provides a useful first-pass pruning threshold. This threshold is determined without training any models: bandwidth selection requires only inexpensive QUESTS $f_{\text{uniq}}(h)$ computations over a range of h values, making the elbow criterion a computationally inexpensive and model-agnostic guide that avoids exhaustive model-based cross-validation. Notably, where the alignment is imperfect the markers err on the conservative side, falling at higher retention than strictly necessary rather than inside the regime of steep error growth. Agreement is weaker in heterogeneous liquid regimes, where flatter or noisier RMSE-retention behavior reduces thresh-

old sharpness. Elbow bandwidths are most reliable when group definitions are physically and structurally coherent.

4 Conclusion

We have developed an adaptive, group-stratified, entropy-guided pruning workflow for MLIP dataset reduction and evaluated it on the multi-phase Carbon ChIMES 2.0 dataset. This method requires only inexpensive QUESTS $f_{\text{uniq}}(h)$ computations and no model training, making bandwidth automated and retention-threshold selection computationally efficient. Using this method, we find that identified bandwidths vary substantially across state points in a physically interpretable way. A single fixed bandwidth therefore gives an incomplete account of within-group similarity. We show that group-stratified pruning improves force and structural accuracy at equal retention for most crystalline and low-density liquid conditions, with the largest improvements under aggressive pruning. These improvements are found to depend on how structurally uniform each group is. High-density liquid regimes with mixed bonding character benefit little, because a single thermodynamic label groups structurally distinct environments together. Extended per-statepoint results (Supplementary Figs. S3 through S6 and S8 through S11) confirm that these improvements are systematic across the full dataset. Better sub-grouping may be achievable through unsupervised descriptor-space clustering, which is the subject of future work. Ultimately, this work provides a robust strategy for building compact MLIP training sets that yield models as accurate as those trained on the full dataset, at a fraction of the labeling and training cost.

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5 Code Availability

All data and code will be made publicly available on GitHub as part of the Orchestrator package

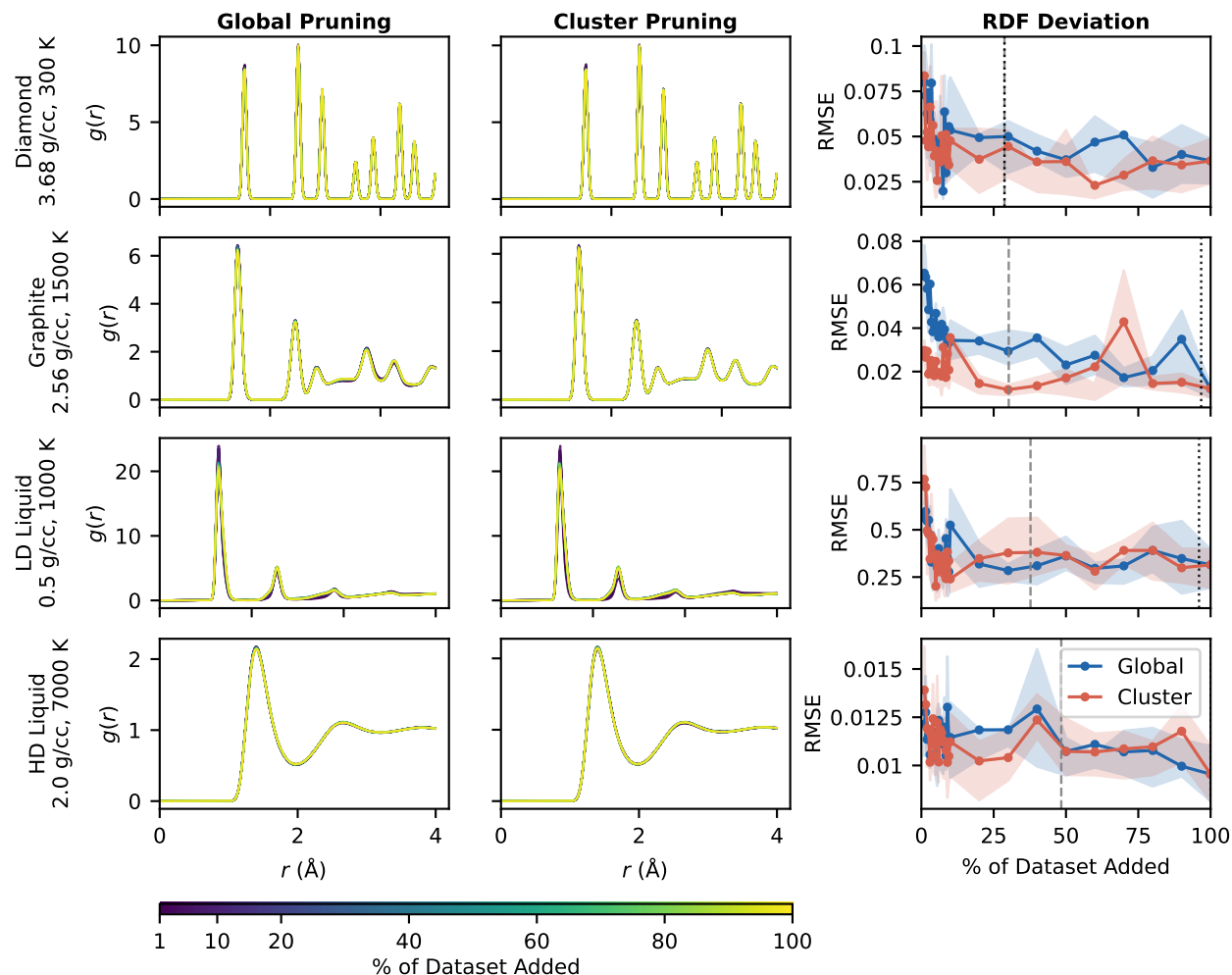


Figure 4: Radial distribution function (RDF) fidelity and RDF-RMSE versus dataset retention for four representative conditions (diamond 3.68 g/cc 300 K, graphite 2.56 g/cc 1500 K, LD liquid 0.5 g/cc 1000 K, HD liquid 2.0 g/cc 7000 K). Left column shows RDFs from Global-Fixed models, middle column shows RDFs from Stratified-Adaptive models, and right column shows RMSE relative to the 100% retention reference. Vertical lines indicate elbow-selected and fixed-bandwidth ($h = 0.015$) retention thresholds. Group-based pruning results in equivalent or improved model performance on predicting this material property.

(<https://github.com/llnl/orchestrator>) upon completion of internal review processes. Prior to this, code can be provided upon reasonable request.

Supporting Information Available

The following files are available free of charge:

- **Supplementary_Information.pdf:** Additional data in support of the findings in this work.

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